S2 Coursework: Coal Mining Accident Analysis

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1. Accident Data

The total number of accidents recorded is $N = 191$ over a period of $L = 40550$ days. The mean rate of accidents is 0.00471 accidents/day or 1.72 accidents/year.

$$\bar{\lambda} = \frac{N}{L} \approx 0.0047 \text{ accidents/day.}$$

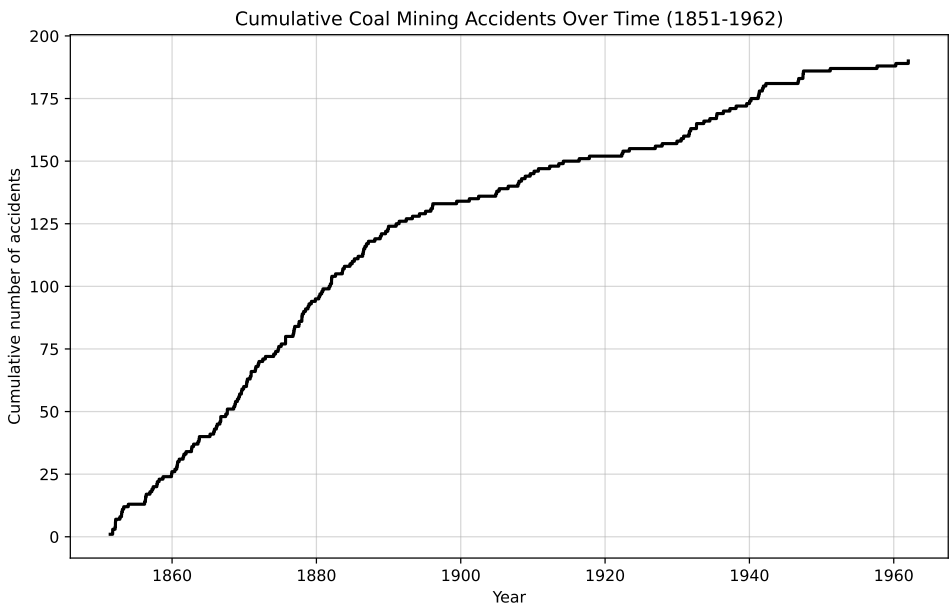

Figure 1: Cumulative number of coal mining accidents from 1851 to 1962. There were 191 accidents recorded over 40550 days.

2. Priors

(a) Qualitative Comparison

The plain order statistics prior involves drawing k change points from a uniform distribution over the interval $[0, L]$ and then sorting them. This means that the points are independent before sorting and can be clustered tightly together. As shown in Figure 2, the marginal distributions are broad and can peak directly at the boundaries.

The even-numbered prior is constructed by drawing $2k + 1$ points uniformly, sorting them, and then taking the evenly indexed points. This results in a repulsive effect between points, encouraging more evenly spaced change points. It also has the effect of preventing peaks at the boundaries, which makes sense for a change point, as the first and last points are always odd

and therefore always excluded. In Figure 2 we can also see that the marginal distributions are more distinctly separated and narrower than those of the plain order priors.

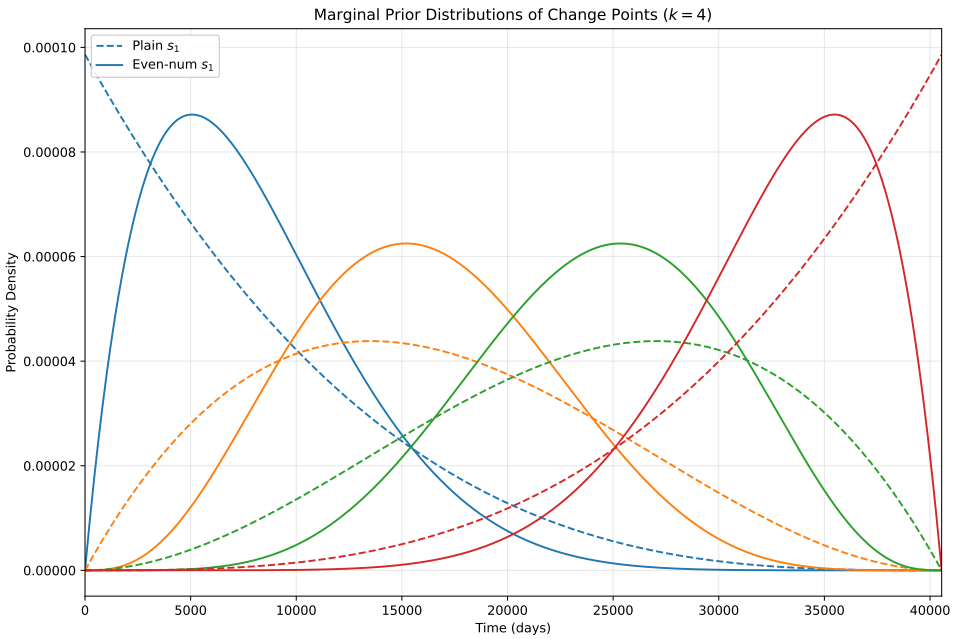

Figure 2: Comparison of plain and even-numbered order statistics priors for $k = 4$ accross different values of j. Plain order priors are shown with a dashed line and even numbered priors with a solid line.

(b) PDF Derivation

To derive the even-numbered pdf we start with the fact that the plain ordered pdf for $2k+1$ points on the unit interval $[0, 1]$ is a constant

$$P(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}) = (2k+1)!$$

This is valid only on the strictly ordered region $0 < u_{(1)} < u_{(2)} < \cdots < u_{(2k+1)} < 1$.

For the even numbered pdf we only keep even-numbered order statistics $\hat{s}_j = u_{(2j)}$ so must marginalise the plain order pdf over all the other discarded points. We know that the points are ordered and on the unit interval $[0, 1]$ so the integral is

$$P(\hat{s}_1, \ldots, \hat{s}_k) = \int_0^{\hat{s}_1} \int_{\hat{s}_1}^{\hat{s}_2} \cdots \int_{\hat{s}_k}^{1} (2k+1)!\, du_{(2k+1)} \ldots du_{(3)} du_{(1)}$$

Because the integrand $(2k+1)!$ is a constant, each integral becomes the length of the interval between the 2 points on the bounds, or the gap $\hat{g}_j$ between them.

$$P(\hat{s}_1, \ldots, \hat{s}_k) = (2k+1)!(\hat{s}_1)(\hat{s}_2 - \hat{s}_1) \ldots (1 - \hat{s}_k)$$

$$= (2k+1)! \prod_{j=1}^{k+1} \hat{g}_j$$

We now need to map this from the unit interval $[0, 1]$ to the actual timeline $[0, L]$. We are given the scaling relationship $s_j = L\hat{s}_j$ so can scale the gaps according to $\hat{g}_j = g_j/L$ and substitute this into the equation

$$P(\hat{s}_1, \ldots, \hat{s}_k) = (2k+1)! \prod_{j=1}^{k+1} \left(\frac{g_j}{L}\right)$$

Since L is a constant it can be pulled out of the product

$$P(\hat{s}_1, \ldots, \hat{s}_k) = \frac{(2k+1)!}{L^{k+1}} \prod_{j=1}^{k+1} (g_j)$$

We must then apply the change of variables formula

$$P(s_1, \ldots, s_k) = P(\hat{s}_1, \ldots, \hat{s}_k) \times |J|$$

where $|J|$ is the Jacobian determinant of the transformation.

Since $s_j = L\hat{s}_j$, the inverse transformation is $\hat{s}_j = \frac{s_j}{L}$. Taking the derivative of each variable gives $\frac{\partial \hat{s}_j}{\partial s_j} = \frac{1}{L}$. The Jacobian determinant is then given by the product of the derivatives and since there are k points,

$$|J| = \left(\frac{1}{L}\right)^k = \frac{1}{L^k}$$

This allows us to obtain the final pdf

$$\pi(s_1, s_2, \ldots, s_k | M_k) = \frac{(2k+1)!}{L^{2k+1}} \prod_{j=1}^{k+1} g_j$$

$$\pi(s_1, s_2, \ldots, s_k | M_k) \propto \prod_{j=1}^{k+1} g_j$$

3. The constant rate model, M_0

From the instructions we know that the log likelihood is given by

$$\log \mathcal{L}(\{I_i\} | \{s_j\}, \{h_j\}, M_k) = \sum_{i=1}^{N} \log h_0 - \int_0^L dt\, x(t)$$

Since $x(t) = h_0$ is a constant over the interval L

$$\log \mathcal{L}(\{I_i\} | h_0, M_0) = N \log(h_0) - h_0 L$$

The likelihood is therefore

$$\mathcal{L}(\{I_i\} | h_0, M_0) = h_0^N \exp(-h_0 L)$$

From Exercise 2, the prior for the height parameters is universally given by a Gamma distribution $h_0 \sim \Gamma(\alpha, \beta)$ where $\alpha = 1$ and $\beta = 200$. This forms the probability density function

$$\pi(h_0 | M_0) = \frac{\beta^\alpha}{\Gamma(\alpha)} h_0^{\alpha-1} \exp(-\beta h_0)$$

Since $\alpha = 1$ and $\Gamma(1) = 1$, the prior simplifies to

$$\pi(h_0|M_0) = \beta \exp(-\beta h_0) = 200 \exp(-200 h_0)$$

Applying Bayes' rule the unnormalised posterior P^* is the product of the prior and likelihood

$$P^*(h_0|\{I_i\}, M_0) = \beta \cdot h_0^N \exp(-h_0(L + \beta))$$

To calculate the true evidence Z_0 for model M_0, we must integrate $P^*(h_0|\{I_i\}, M_0)$ over all possible values of h_0.

$$Z_0 = \int_0^\infty P^*(h_0|\{I_i\}, M_0)\, dh_0$$

When evaluating $P^*(h_0) = \beta \cdot h_0^N \exp(-h_0(L+\beta))$ for our dataset where $N = 191$ and $L = 40550$, the probabilities evaluate to 0 due to an effect known as floating point underflow.

To safely resolve this underflow, we perform the computation in log-space. First, we take the logarithm of the unnormalised posterior:

$$f(h_0) = \log(P^*(h_0)) = \log(\beta) + N \log(h_0) - h_0(L + \beta)$$

We can then find the exact maximum of this log-posterior analytically. Taking the derivative with respect to h_0 and setting to zero yields the maximum a posteriori (MAP) height:

$$h_{\max} = \frac{N}{L + \beta} \approx 0.00469$$

By evaluating the maximum log-posterior $C = f(h_{\max})$, we can shift the posterior to peak at 1 and prevent the underflow problem when calculating the evidence.

$$P^*(h_0) = \exp(C) \times \exp(f(h_0) - C)$$

$$Z_0 = \exp(C) \int_0^\infty \exp(f(h_0) - C)\, dh_0$$

We can now use numerical integration with `scipy.integrate.quad` to evaluate the integral

$$\log Z_0 = C + \log\left(\int_0^\infty \exp(f(h_0) - C)\, dh_0\right) \approx -1217$$

The prior and posterior distributions are plotted in Figure 3.

4. The 1 change point model, M_1

(a) MCMC Analysis

For the MCMC analysis, we use the `emcee` package which implements an affine invariant ensemble sampler. It works by creating an ensemble of multiple walkers that propose their next moves based on the position of the other walkers in the ensemble. This makes them invariant to the scale of the parameters, which is important given that s_1 is in the order of 10^4 and h_0,h_1 are in the order of 10^{-3}.

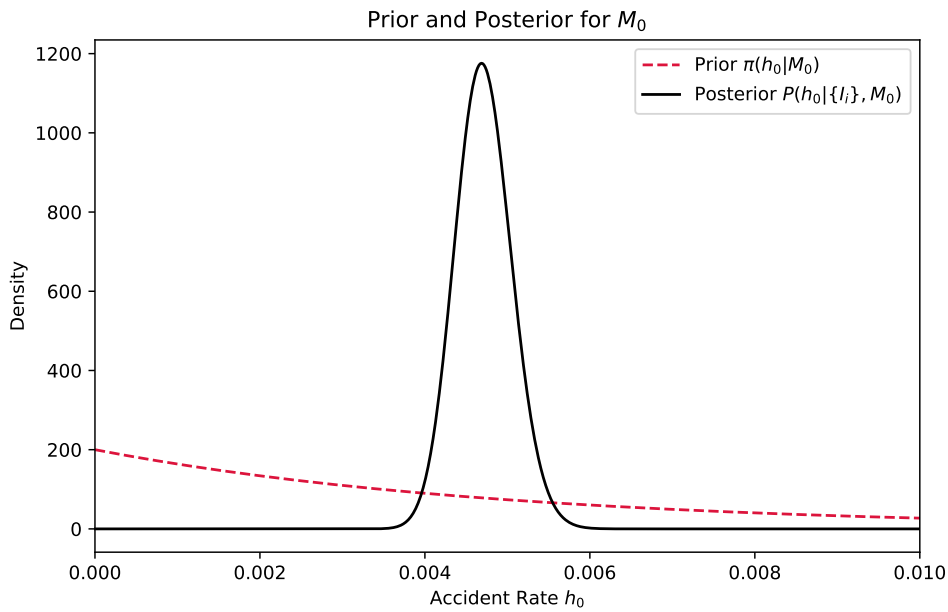

Figure 3: Prior (dashed line) and posterior (solid line) distributions for the constant rate model M_0. The maximum a posteriori value for the accident rate h_0 is approximately 0.0047 accidents per day.

There are 3 dimensions to search over $[s_1, h_0, h_1]$ so 32 walkers were used to ensure full coverage. From the trace plots in Figure 4 we can see that the burn-in period for the chains was very short and that the set value of 500 steps was sufficient to allow the chains to converge. This fast convergence is likely due to the fact that the walkers were initialised at values close to the peaks for each parameter, chosen from the results of M_0. The trace plots each remain stable within a flat, horizontal band suggesting that they reached a stationary distribution. The autocorrelation times (τ) for the chains were calculated to be $44 < \tau < 52$ meaning that the 8000 steps were easily enough to obtain a large number of independent samples. The chains were not thinned as the sample size is too small to be concerned with memory usage. The Gelman-Rubin statistic was not calculated as the walkers look at other walkers to determine their next step, meaning they are inherently not independent.

The corner plot in Figure 5 shows that the s_1 parameter has 3 peaks and that a single change point may not accurately describe the data.

(b) Savage Dickey Density Ratio

The Savage-Dickey Density Ratio (SDDR) is a method for calculating the evidence ratio of two nested models with consistent priors. Since M_0 can be interpreted as a special case of M_1 where $h_0 = h_1$, the models are nested and the priors on the parameters h are consistent as they given by the same Gamma distribution. This means that the SDDR can, in principle, be used to find the evidence ratio of M_1 to M_0.

We calculate the SDDR by the ratio of the posterior and prior density under the nested condition $\Delta h = h_0 - h_1 = 0$.

$$B_{01} = \frac{P(\Delta h = 0|D, M_1)}{\pi(\Delta h = 0|M_1)}$$

The prior density can be calculated analytically

$$\pi(\Delta h) = \frac{\beta}{2}\exp(-\beta|\Delta h|) = \frac{200}{2}\exp(0) = 100$$

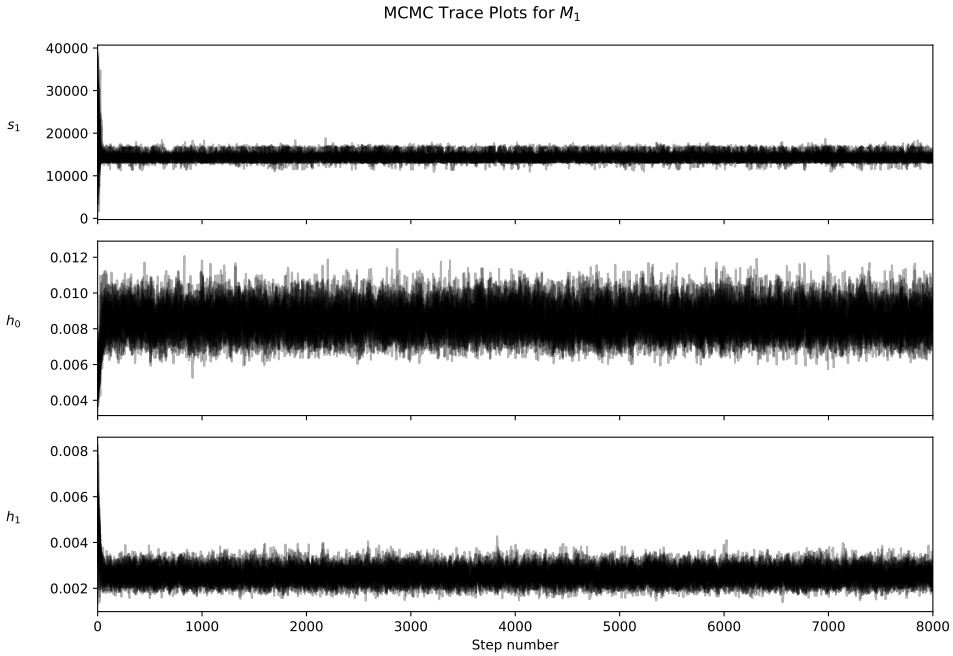

Figure 4: Trace plots from the ensemble MCMC sampler for the 1-change point model, M_1. The three panels show the evolution of the 32 walkers over 8000 steps for the parameters s_1, h_0, and h_1.

The problem arises in this example when trying to evaluate the posterior density at $\Delta h = 0$. From Figure 5, we can see that the marginalised posteriors for h_0 and h_1 are tightly constrained around 0.0085 and 0.0026 respectively, meaning there will be very little overlap and very few, if any, MCMC samples with $\Delta h = 0$. As a result, we can see that the kernel density estimate for the posterior difference in Figure 6 peaks at ≈ 0.006 and has effectively 0 density at $\Delta h = 0$. This will result in an SDDR B_{01} of 0 and $\log B_{01} \to -\infty$.

(c) Laplace Approximation

The Laplace approximation is a semi analytic technique for calculating the evidence which approximates the posterior as a Gaussian around its peak. For M_1 there are two issues that prevent us from using this approximation effectively.

$$\log Z \approx \log P(D|\theta_{MAP}, M) + \log \pi(\theta_{MAP}|M) + \frac{d}{2}\log(2\pi) - \frac{1}{2}\log|H|$$

The main issue is that the Laplace approximation requires the posterior to be smooth as it calculates the Hessian matrix H. As we move along s_1, the likelihood is completely flat until a day where an accident occurs, at which point it discontinuously jumps. This will result in an undefined derivative at data points and 0 everywhere else. The determinant of the Hessian $|H|$ will therefore equal 0 and the $-\frac{1}{2}\log(0)$ term will tend to infinity.

The second issue is that the posterior is multimodal, as can be seen in Figure 5 where the marginal posterior for s_1 has 3 distinct peaks. Fitting a single Gaussian around the main peak would ignore an significant proportion of the distribution, leading to an underestimation even if the Hessian was well defined.

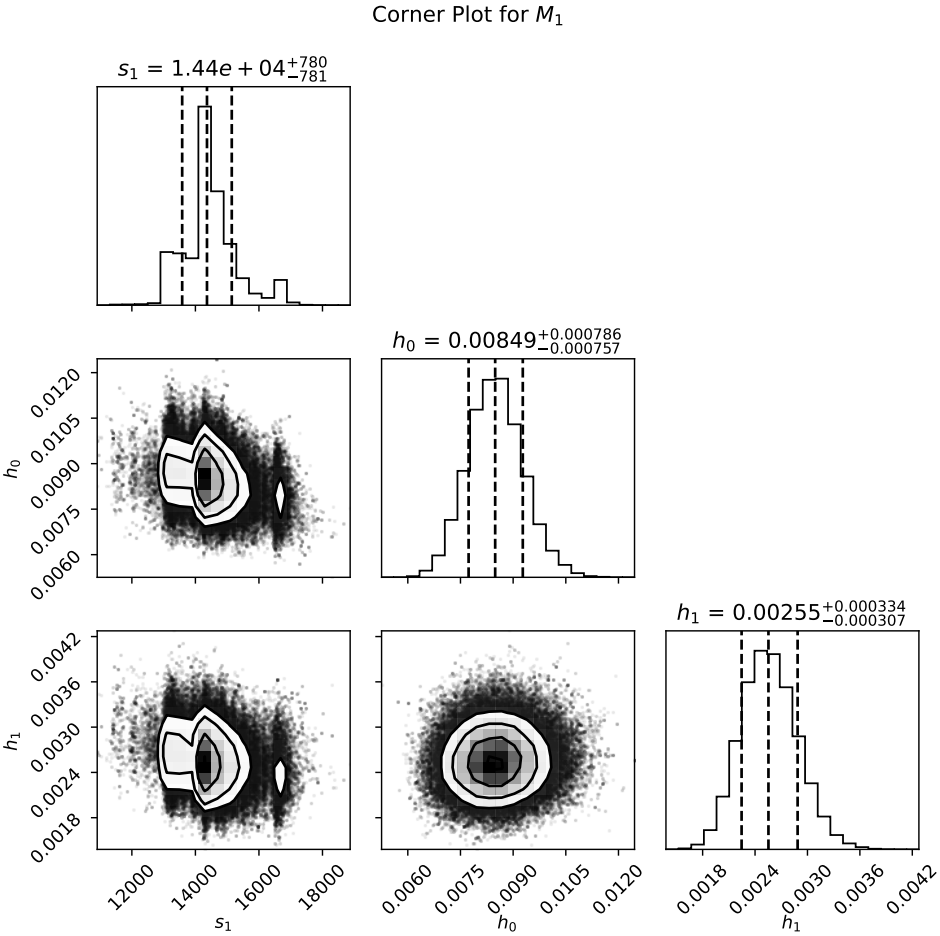

Figure 5: Corner plot showing the 1D and 2D marginal posterior distributions for the parameters s_1, h_0, and h_1 in the 1-change point model, M_1. The diagonal panels display the 1D marginal posteriors for each parameter, with dashed vertical lines indicating the 16th, 50th (median), and 84th percentiles.

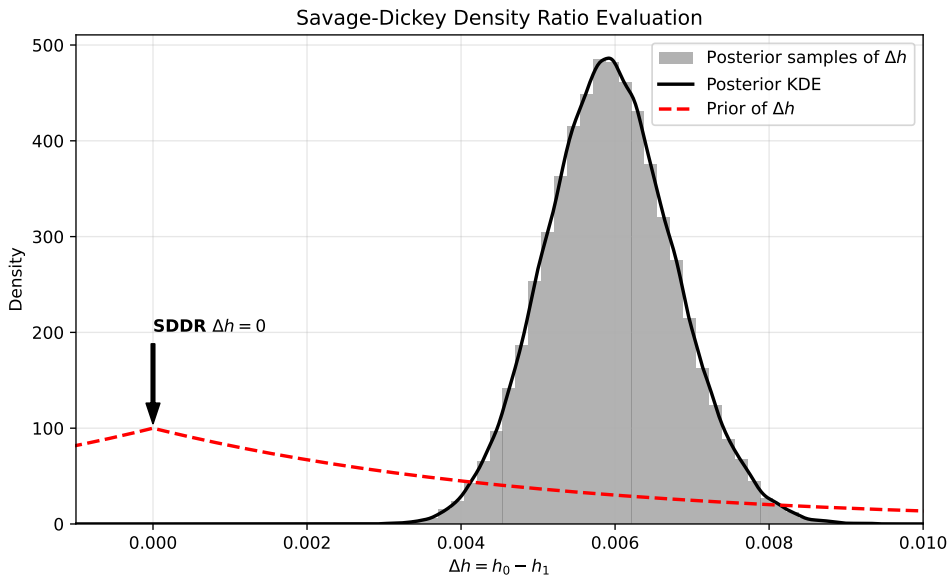

Figure 6: The plot displays the prior density (dashed red line), the posterior samples (grey histogram), and the posterior Kernel Density Estimate (solid black line) for the difference in accident rates, $\Delta h = h_0 - h_1$. The SDDR is evaluated at the nested condition $\Delta h = 0$, marked by the arrow. While the analytical prior density at this point is 100, the posterior density is effectively zero, with the distribution peaking around 0.006.

(d) Nested Sampling

To calculate the evidence Z we use the nested sampling package `dynesty`. Nested sampling works by initialising a set of live points and then iteratively replacing the points of lowest likelihood with one of higher likelihood. By keeping track of how the volume of the prior space changes the algorithm can numerically integrate the posterior to find Z.

We used 1000 live points to ensure a high resolution for numerical integration with a default stopping criterion of $\delta \log Z = 0.1$. Design choices to improve the efficiency of the algorithm included setting `bound='multi'` rather than the default single ellipsoid. The use of multiple bounds is important as we have seen in Figure 5 that the posterior is multimodal. Another choice was to use `sample='rwalk'` method for sampling, which is well-suited to lower dimensional multimodal problems.

The calculated evidence for M_1 was $\log Z_1 = -118.53 \pm 0.10$ resulting in a log Bayes factor $\log B_{10} \approx 28.56$ and the Log Posterior Odds ≈ 29.66. This suggests that M_1 is mathematically superior to M_0.

5. The k change point models, M_k

(b) MAP Change Points

The RJMCMC was run for 150,000 iterations, with the results shown in Figure 7. From the trace plot we can see that the model explored the space up to $k = 7$ and the running average stayed roughly between 1 and 2 suggesting the chain converged. The posterior distribution peaks at $k = 1$ which is the maximum a posteriori estimate.

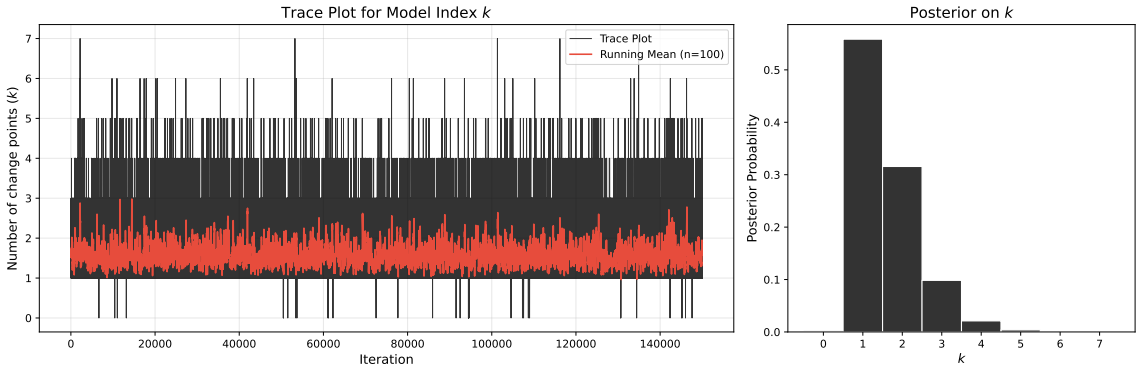

Figure 7: Reversible Jump MCMC (RJMCMC) results for the model index k over 150,000 iterations. **Left:** Trace plot of the number of change points, demonstrating that the chain successfully explores the model space up to $k = 7$. The red line shows the running mean ($n = 100$), which remains stable between 1 and 2, indicating convergence. **Right:** The marginal posterior probability distribution for k. The distribution peaks at $k = 1$, indicating the maximum a posteriori estimate.

(c) Model Averaging

To calculate the uncertainty regions, we first create an array of 500 evenly spaced points between 0 and L. We then remove the first 10% of RJMCMC samples to account for burn in and for each sample in the remaining 'clean' chain we evaluate $x(t)$ for each point in the time array. The result is a $135,000 \times 500$ matrix where each row is an MCMC iteration and every column is a specific point in time.

The RJMCMC samples models in proportion to their posterior so evaluating the distribution of $x(t)$ values at each time step (down the column) naturally marginalises out the number of change points k. We can then construct the median rate and the 50% and 90% credible intervals shown in Figure 8 by calculating the 5th, 25th, 50th, 75th, and 95th percentiles at each time point.

AI Use

I have used autogeneration tools to assist with code debugging, LaTeX formatting and helping to generate figures. This declaration is made in accordance with the Course Handbook.

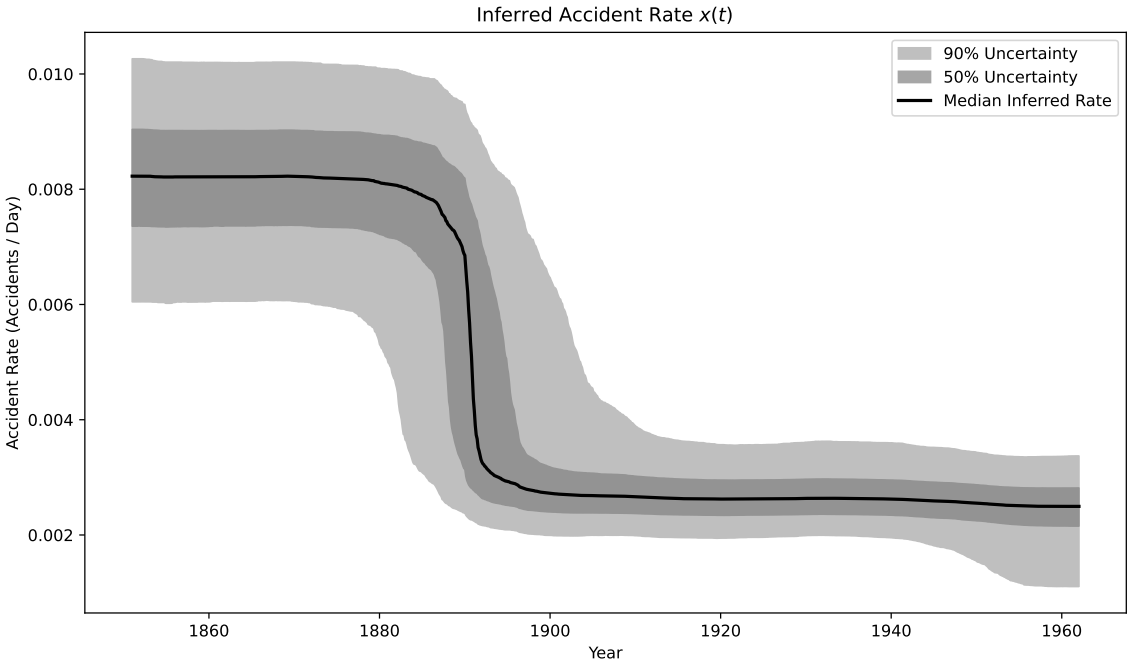

Figure 8: Inferred accident rate $x(t)$ over time, marginalised over the number of change points k using model averaging from the RJMCMC samples. The solid black line represents the median inferred rate. The dark and light grey shaded regions indicate the 50% and 90% credible intervals respectively, calculated from the percentiles of the posterior samples at each time step.